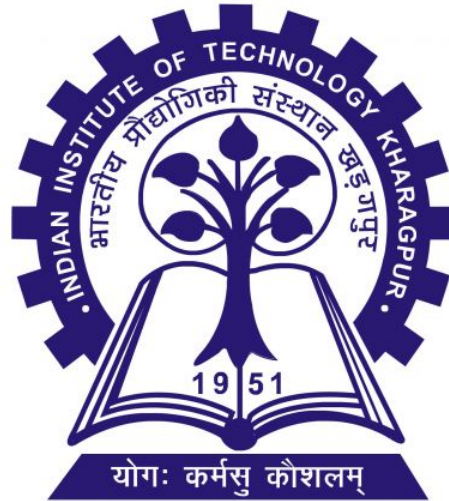


VLSI DESIGN LAB [EC69216]



EXPERIMENT - 6

Design and Layout of Current Reference Circuit

Group No. 4

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- **Group no.:** 4
- **Date of Submission:** 30-03-2026

1 Aim of the Experiment

The objectives of this experiment are:

1. To design and simulate a current reference circuit providing a constant current of 100 at typical process corners, with $V_{dd} = 1.8$ and $T = 300$.
2. To ensure the circuit operates correctly down to a minimum supply voltage of $V_{dd(min)} = |V_{th}| + 0.5$.
3. To incorporate a necessary start-up circuit to prevent the reference from settling into a zero-current degenerate state.
4. To size the transistors appropriately such that they comply with common-centroid layout requirements.
5. To perform a DRC and LVS clean physical design (layout) of the current reference and verify its post-layout performance.

2 Introduction

Analog circuits heavily rely on stable biasing networks to establish proper operating points for amplifiers, oscillators, and data converters. A robust current reference is expected to provide a stable DC current that exhibits minimal sensitivity to variations in power supply voltage (VDD), temperature, and process parameters (PVT variations).

Simple current mirrors tied to VDD fail to meet these criteria because their output current scales directly with the supply voltage. To achieve supply independence, a self-biased architecture, often referred to as a Beta-Multiplier Current Reference (BMCR) or Constant- G_m bias circuit, is widely used. This architecture employs positive feedback with loop gain less than unity to stabilize the current, relying on a physical resistor to dictate the reference current value.

3 Theory

3.1 Beta-Multiplier Current Reference Core

The core of the Beta-Multiplier Current Reference consists of a PMOS current mirror that forces equal currents through two NMOS branches. One of the NMOS transistors is source-degenerated using a resistor (R_s). Furthermore, the NMOS transistor with the source resistor is sized wider than the other NMOS transistor by a factor of K (where $K > 1$).

Because the PMOS mirror forces $I_{D1} = I_{D2} = I_{REF}$, and the gate voltages of both NMOS transistors are tied together, the voltage across the resistor is equal to the difference in the gate-source voltages of the two NMOS devices:

$$I_{REF} \cdot R_s = V_{GS1} - V_{GS2}$$

Assuming square-law operation in strong inversion, the reference current can be derived as:

$$I_{REF} = \frac{2}{\mu_n C_{ox} (W/L)_1 R_s^2} \left(1 - \frac{1}{\sqrt{K}} \right)^2$$

This expression reveals that, to a first order, I_{REF} is independent of the supply voltage V_{dd} . It is primarily determined by the physical resistor R_s , the sizing ratio K , and process parameters ($\mu_n C_{ox}$).

3.2 Minimum Supply Voltage ($V_{dd(min)}$)

For the beta-multiplier circuit to operate correctly, all transistors must remain in the saturation region. The critical path for the minimum supply voltage typically includes the diode-connected PMOS and the NMOS without source degeneration. The minimum supply voltage required is the sum of the gate-source voltage of the PMOS and the drain-source saturation voltage of the NMOS:

$$V_{dd(min)} = |V_{GS,p}| + V_{DSAT,n} \approx |V_{th,p}| + |V_{ov,p}| + V_{ov,n}$$

According to the design specifications, the overdrive voltages (V_{ov}) must be carefully managed through transistor sizing to ensure the circuit functions properly at $V_{dd(min)} = |V_{th}| + 0.5$.

3.3 Start-up Circuit

Self-biased circuits like the BMCR possess two stable operating points. The desired state provides the calculated I_{REF} , but a secondary, degenerate state exists where zero current flows ($I_{REF} = 0$), driving all transistor gate-source voltages to zero.

To prevent the circuit from remaining in this degenerate state upon power-up, a start-up circuit is mandatory. The start-up circuit monitors the current in the reference loop. If the current is zero, the start-up network injects a small initial current into the loop to trigger operation. Once the core circuit powers up and reaches its intended steady-state operation, the start-up circuit automatically turns off to avoid interfering with the reference current stability.

3.4 Sizing for Common Centroid Layout

To mitigate the effects of process gradients (e.g., oxide thickness variations, doping concentration gradients) across the silicon wafer, critical matched transistor pairs (like the PMOS current mirrors and the NMOS differential pair) must be laid out using common centroid topologies.

For a common centroid layout to be feasible, the transistor widths must be partitioned into identical unit-sized fingers (M multipliers). The design ratio K must ideally be an integer that allows symmetric placement around a central axis (e.g., placing the unit device in the center and the K -sized device's fingers symmetrically around it, or using an A-B-B-A 1D/2D interdigitation pattern). Proper choice of unit width (W_{unit}) and finger length (L) during the schematic phase is vital for DRC/LVS clean and symmetric physical routing.

4 Experimental Results

4.1 Device Characterization

Before designing the core beta-multiplier circuit, individual NMOS and PMOS devices were characterized to determine their threshold voltages and the K_n' and K_p' values to be used in current determining equation. This step is crucial for accurate hand calculations and initial sizing.

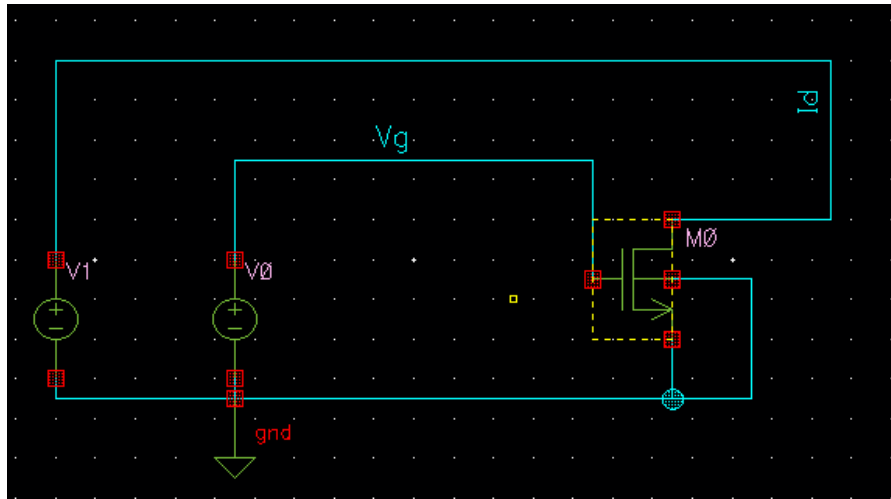


Figure 1: Testbench circuit used for device characterization.

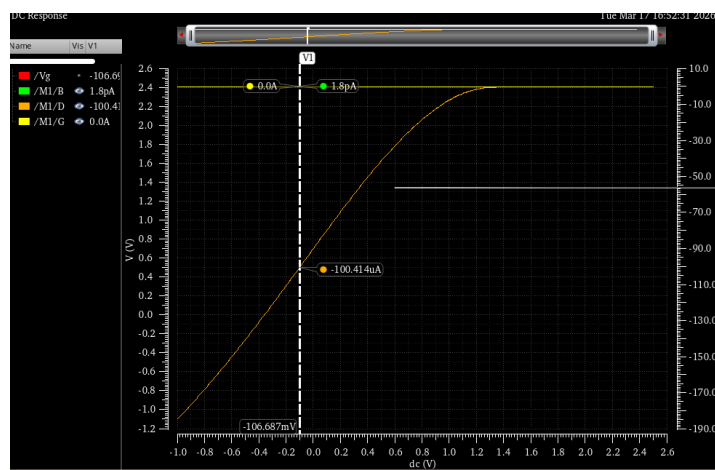
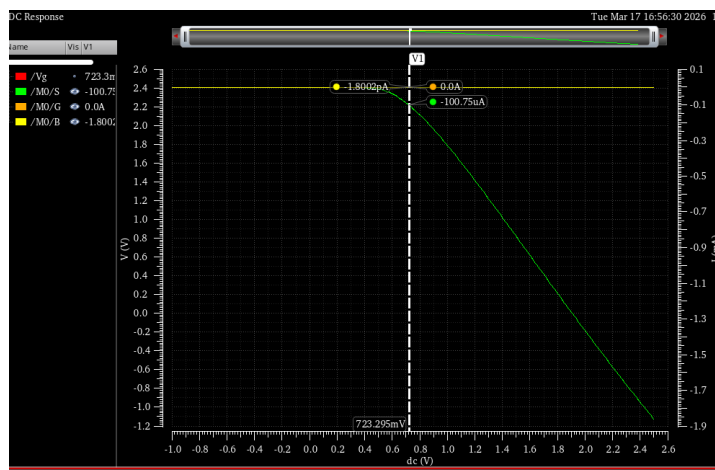


Figure 2: NMOS DC characteristic (I_d vs V_g).

Figure 3: PMOS DC characteristic (I_d vs V_g).

From these simulations at the target current $100\text{ }\mu\text{A}$, the values were calculated (using square-law at saturation region) and observed from the i-v characteristics graph and model parameters. They were found to be:

- Kn': $187\mu\text{A}/\text{v}^2$
- Kp': $57\mu\text{A}/\text{v}^2$

4.2 Schematic Design and Pre-Layout Simulation

The schematic of the beta-multiplier current reference, including the start-up circuit, was drafted and sized to achieve the target $100\mu\text{A}$ current.

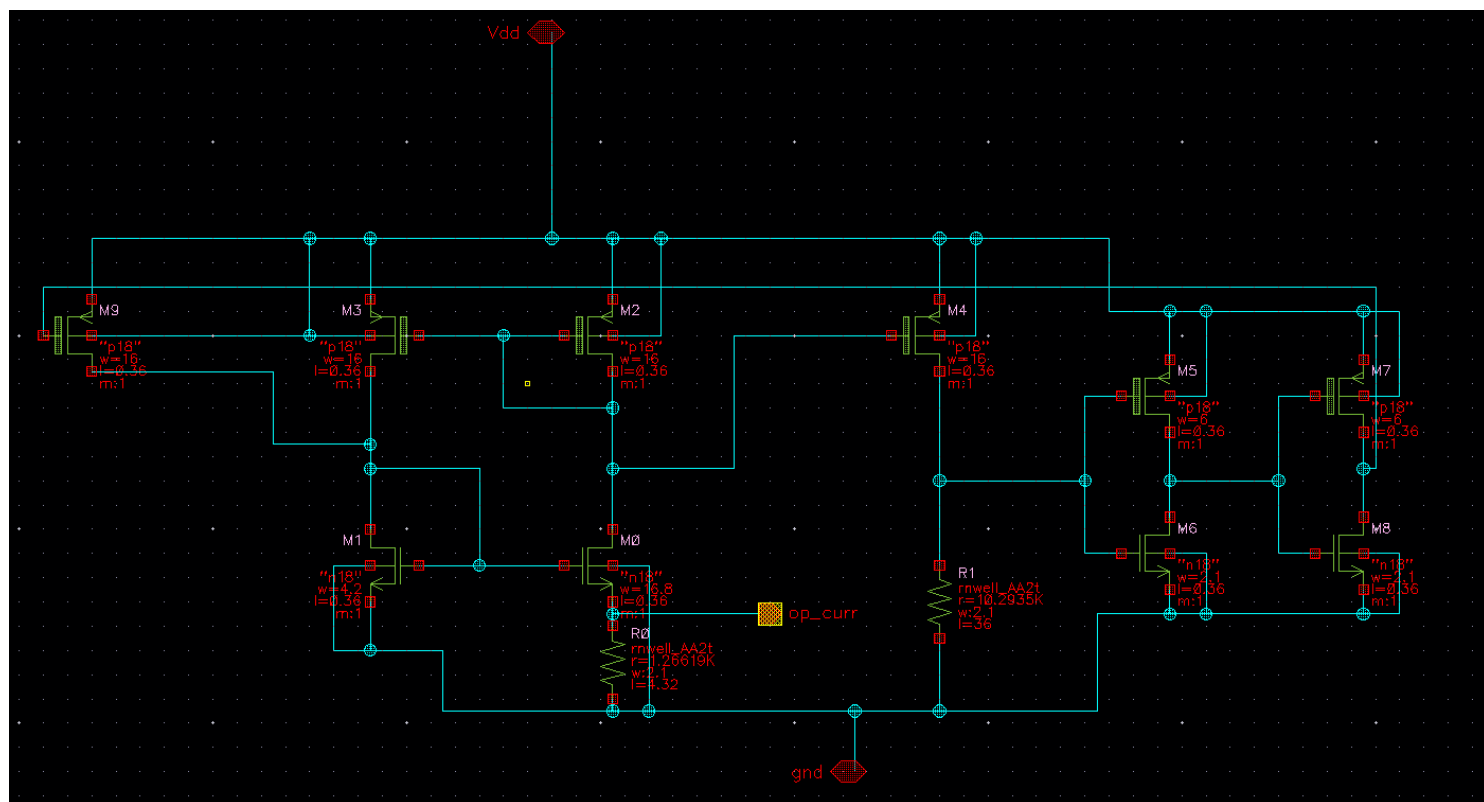


Figure 4: Complete schematic of the current reference circuit including the start-up network.

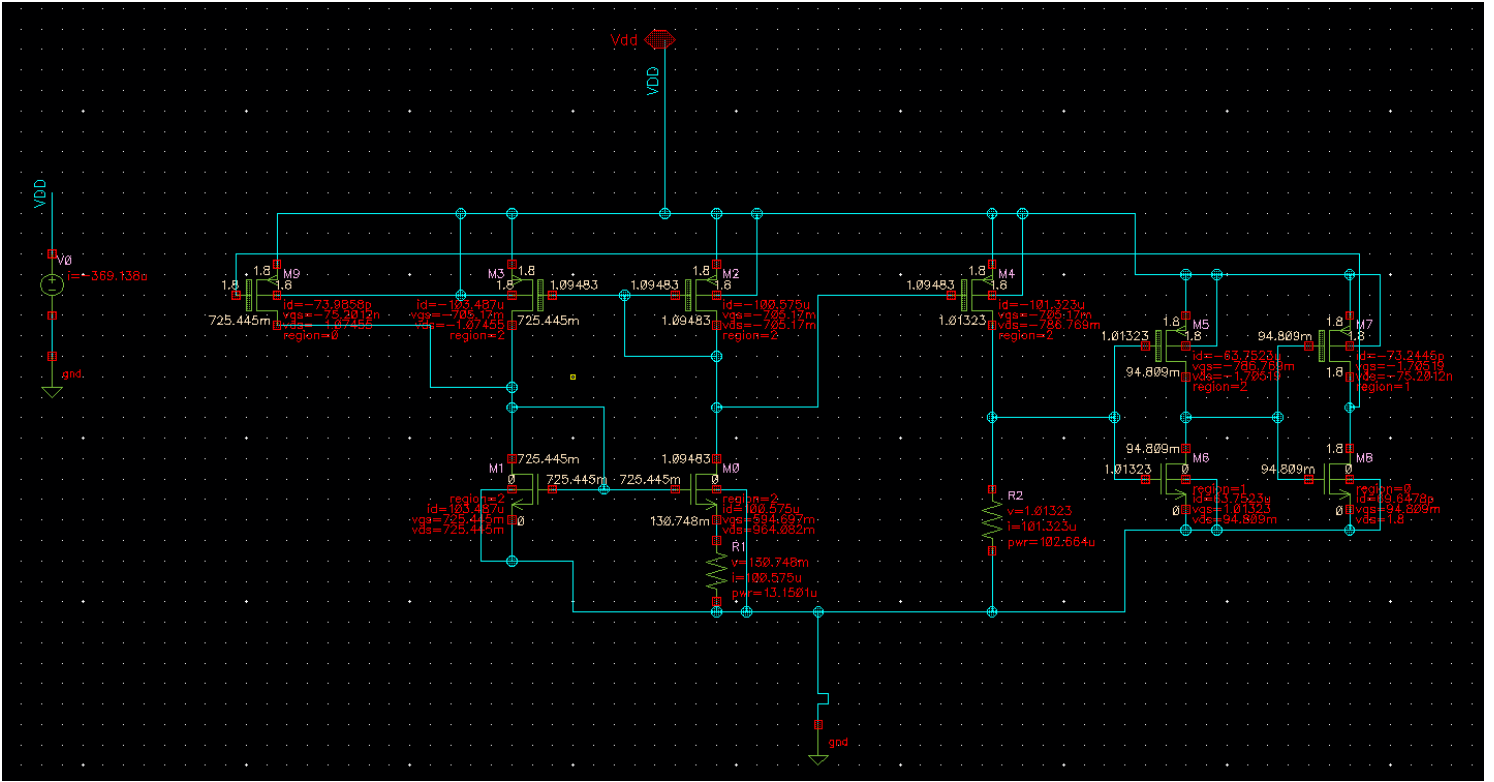


Figure 5: DC operating point annotation of the schematic confirming the regional operation of the transistors.

Pre-layout transient simulations were executed to verify the stability and value of the reference current and operational voltage over time. The schematic-level simulation yielded an output current of approximately $99.73\mu\text{A}$, which is extremely close to the required $100\mu\text{A}$ specification.

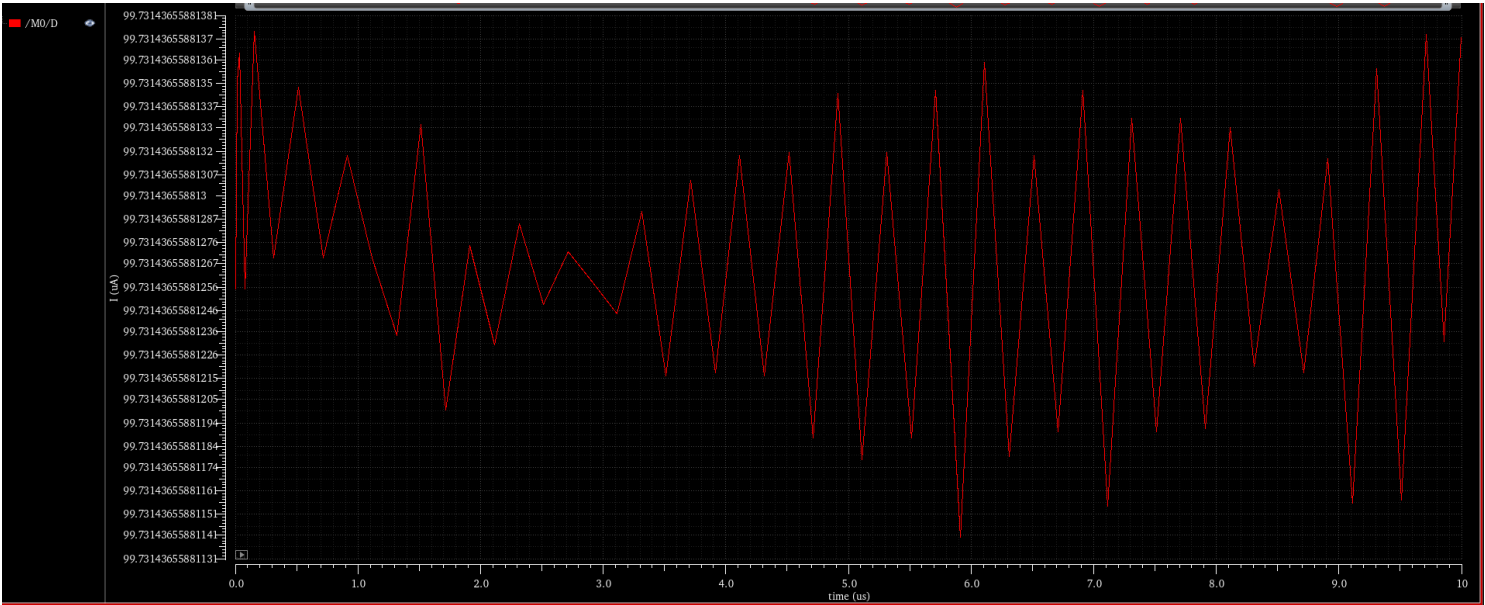


Figure 6: Pre-layout transient simulation of the reference current, settling at approx $99.7\mu\text{A}$.

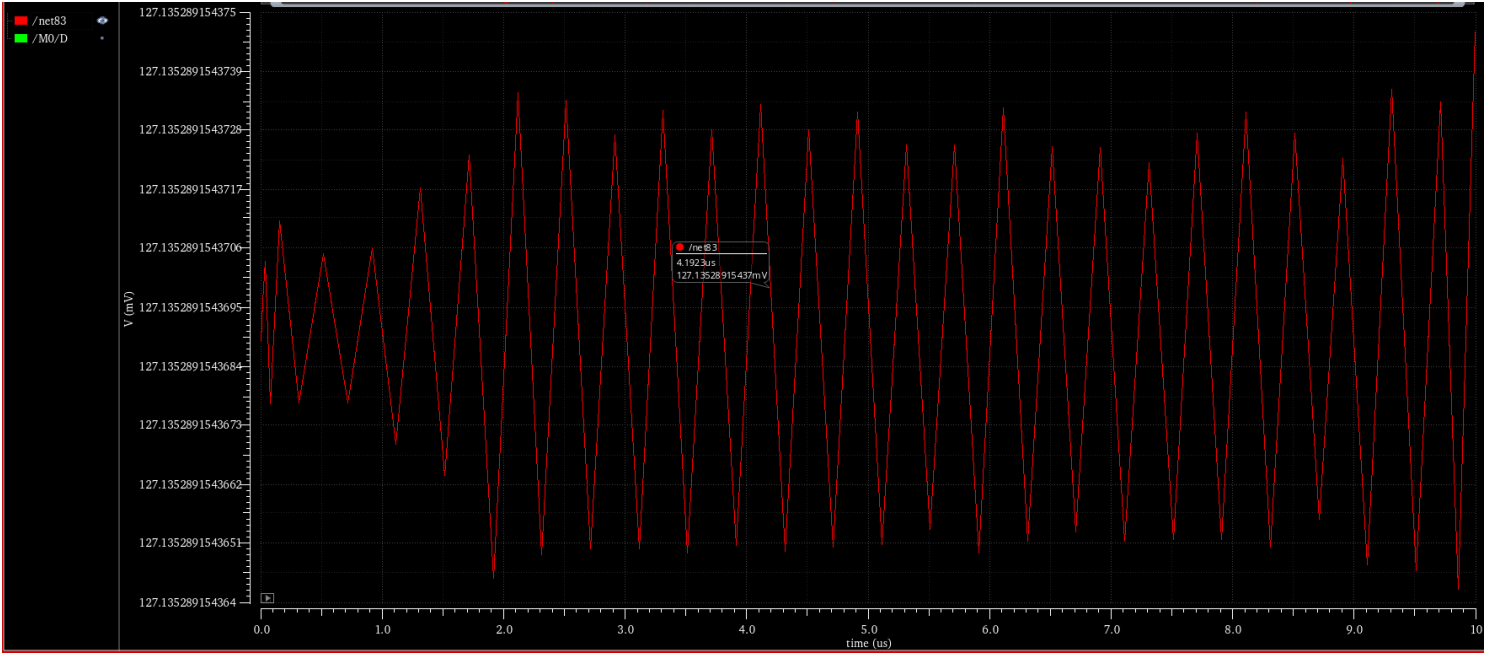


Figure 7: Pre-layout transient simulation of the operating voltage.

4.3 Physical Design (Layout) and Verification

The physical layout was designed strictly adhering to common centroid matching techniques for both the PMOS current mirrors and the NMOS differential pairs to minimize the impact of process variations. Dummy transistors were utilized at the edges of the matched arrays to prevent boundary effects during fabrication.

4.4 Post-Layout Simulation and Comparison

Following RC extraction, post-layout simulations were conducted to observe the impact of parasitic capacitances and resistances introduced by the physical routing.

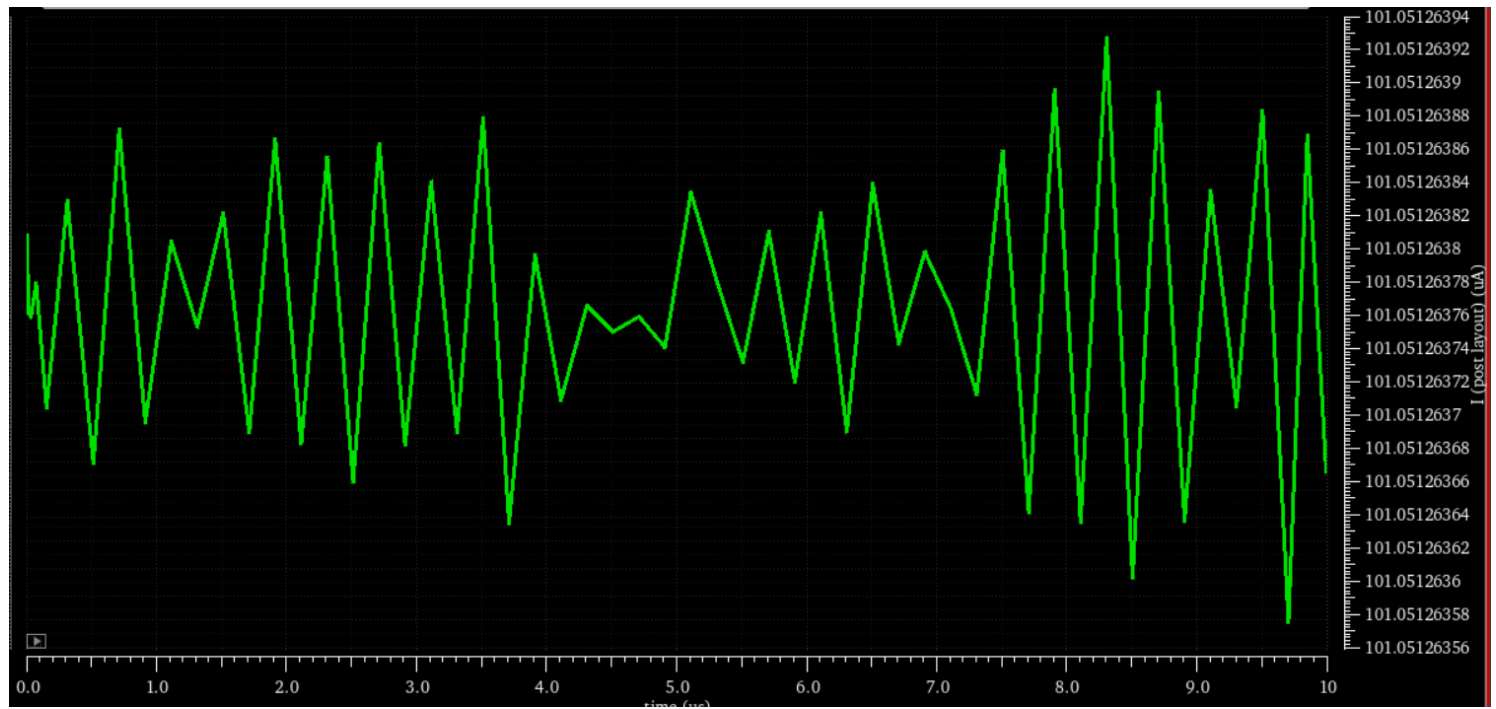


Figure 11: Post-layout transient simulation of the reference current, settling at $101.05\mu\text{A}$.

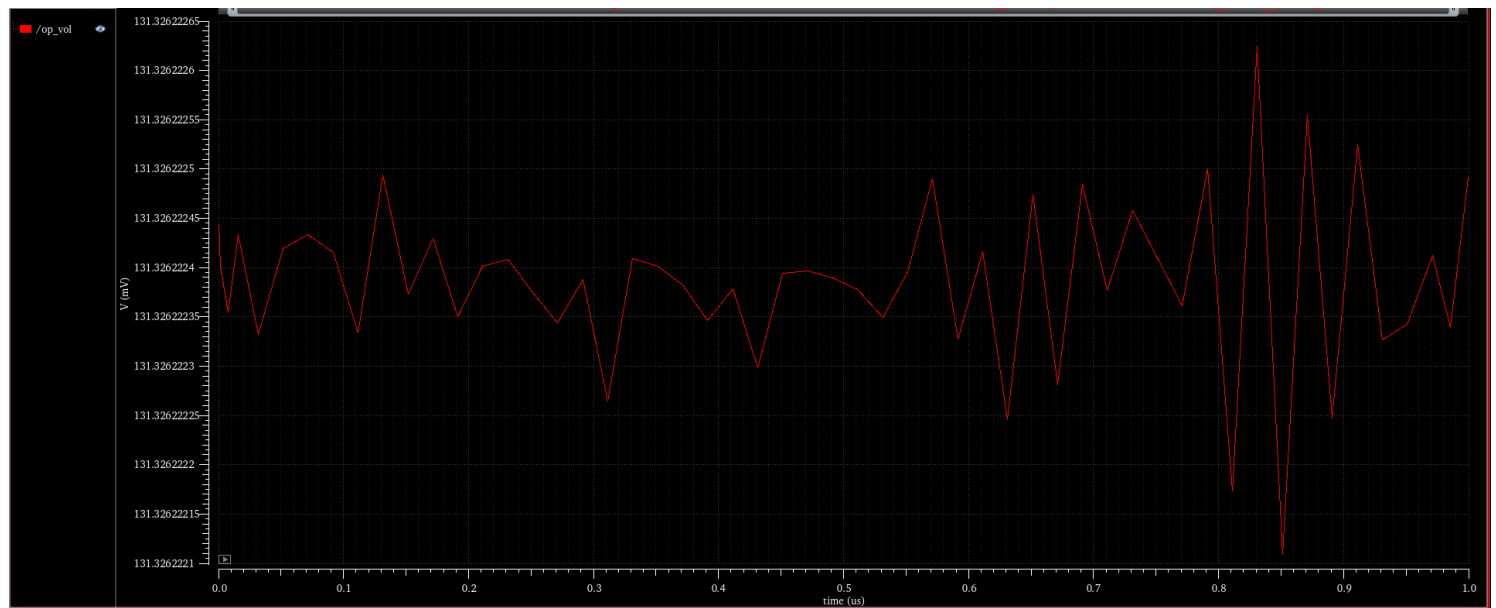


Figure 12: Post-layout transient simulation of the operating voltage.

To clearly visualize the effect of layout parasitics, the pre-layout and post-layout transient current waveforms were superimposed. Due to the added parasitics, the post-layout current shifted slightly to approximately $101.05\mu\text{A}$, which remains well within acceptable tolerance limits for a $100\mu\text{A}$ target.

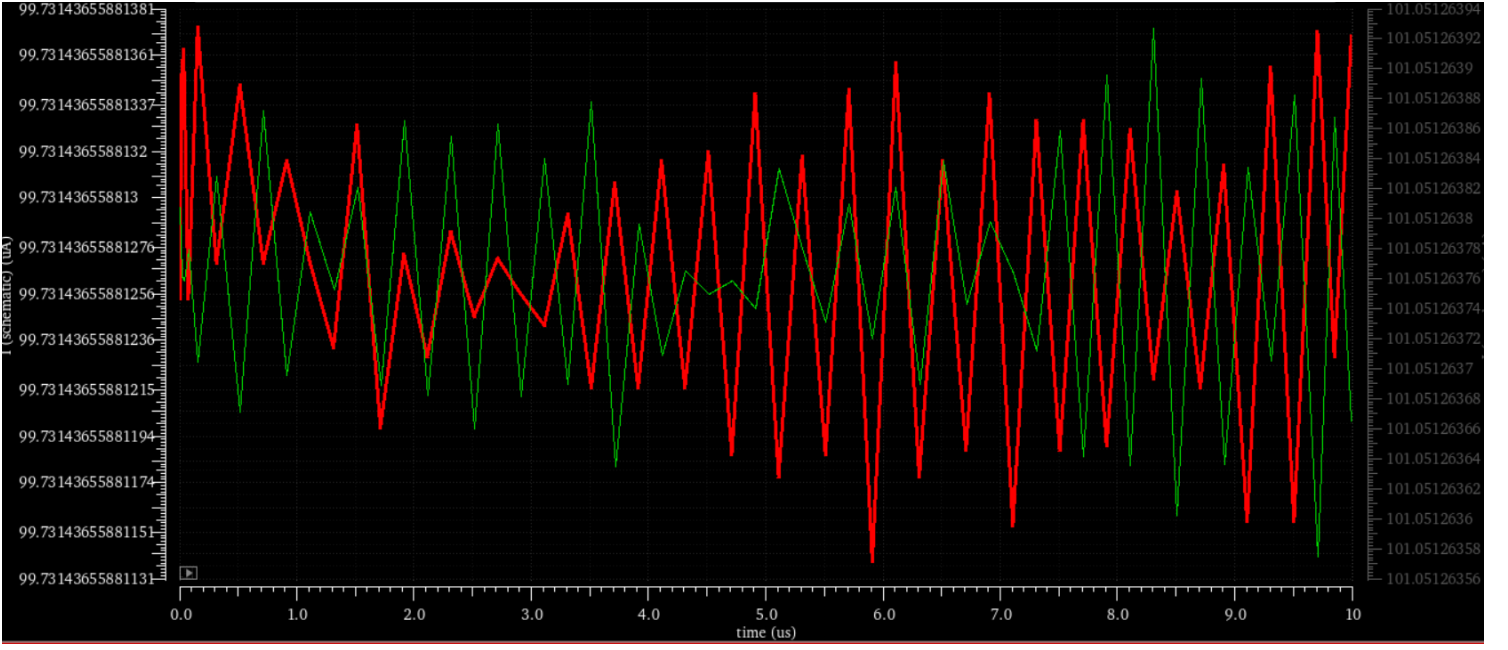


Figure 13: Superimposed waveforms of pre-layout (red) and post-layout (green) reference currents.

5 Discussion Arup Basak 22EC37021

Based on the experimental procedure and the obtained pre-layout and post-layout results, the following key observations and technical points can be discussed:

- The pre-layout transient simulation yielded a steady-state reference current of approximately 99.73. This is exceptionally close to the target specification of 100. The minor discrepancy (less than 0.3%) can be attributed to second-order effects in the advanced BSIM device models used by the simulator (such as channel-length modulation and mobility degradation), which are typically simplified during initial hand calculations.
- The post-layout simulation revealed a slight increase in the reference current to approximately 101.05. As clearly observed in the superimposed waveform plot, the physical routing introduces parasitic resistances (from metal interconnects and vias) and parasitic capacitances. These parasitics slightly alter the effective resistance in the source-degenerated NMOS branch and the bias points of the current mirrors, leading to the observed ~ 1.3 shift. However, this shift is minimal and well within acceptable design tolerances for an analog reference.
- To ensure robustness against process variations (such as oxide thickness and doping concentration gradients across the die), a common centroid layout was meticulously implemented for the matched transistor pairs. Distributing the transistor fingers symmetrically ensures that any linear process gradient affects both devices in a matched pair equally, thereby preserving the critical current mirror ratios (1 : 1) and the K -multiplier ratio required for the Beta-multiplier core.
- The transient responses confirm the successful operation of the start-up circuit. Initially, the circuit forces current into the reference loop to prevent the system from settling into the zero-current degenerate state. Once the reference current approaches its nominal value, the start-up circuit properly disengages, preventing any steady-state power dissipation or interference with the core reference loop.
- The device sizing was carefully optimized to maintain all transistors in the saturation region while keeping overdrive voltages (V_{ov}) low. This careful sizing ensures that the circuit can continue to operate and maintain the reference current even if the supply voltage drops to the required $V_{dd(min)} = |V_{th}| + 0.5$, maximizing the usable dynamic range of the design.

6 Discussions Chiradip Biswas 22EC37004

- In this experiment we designed a current reference circuit using the closed loop current mirror topology, achieving a current of $99.73 \mu\text{A}$ (pre-layout), $101.5 \mu\text{A}$ (post-layout), with target as $100 \mu\text{A}$.
- The following equation was used for hand calculation of the reference circuit resistance value:

$$I_{REF} = \frac{2}{\mu_n C_{ox}(W/L)_1 R_s^2} \left(1 - \frac{1}{\sqrt{K}}\right)^2$$

The N was set as 4. The $\mu_n C_{ox}$ term was found from nMOS characterization.

- The main current reference circuit requires an additional start up circuit, as it has 2 stable operating points, one of which is 0 current one. For this purpose the nmos with resistor in series (a current to voltage converter) and 2 cascaded inverters(isolater) and the additional pmos with it's drain common to the lower nMOS of main reference circuit (voltage to current converter, injecting current in the loop), were used.
- Layout of the same circuit was made, in common-centroid manner. This technique is essential while making 2 matching parameter devices, in a fabrication environment where process variation is unavoidable.
- For common centroid layout, all the devices in the schematic were made using even number of fingers, so that each can be placed in mirror symmetry manner. The total width: W, finger number: N_f , finger width: W_f

$$W_f = \frac{W}{N_f} \quad (1)$$

- The resistors were fabricated using series n-well regions, adjusting the widths of each. As evident from the layout, the 2 resistors take almost same space as all the transistors in the circuit. So better practice is not to directly use resistor, rather using active load.
- Although the circuit made, gives close to $100 \mu\text{A}$, when we will apply load, it will undergo loading and this current will deviate significantly. So a more effective technique to obtain the current outside the circuit (considering it as a symbol), is to use cascode current mirroring, as it provides high resistance (good for an ideal current source).